

AUTONOMOUS SOFTWARE EXPERIMENTS ON HERA: ANNEX D - BIN2C.PY, BINARY TO C HEADER CONVERTER

Reference	ESA-HERA-TECS-MAN-2026-002053
Issue/Revision	1.0
Date of Issue	19/06/2026



Table of Contents

1. Overview 3

2. Prerequisites 3

3. How to Use the Script 3

3.1. Command Line Syntax 3

3.2. Example Usage 3

3.3. Output 3

3.4. Understanding the Generated Header (images_data.h) 3

3.5. File Structure Example 4

1. OVERVIEW

`bin2c.py` is a Python tool designed for testing AFC images. Its primary purpose is to embed raw AFC image files directly into a C application.

2. PREREQUISITES

- Python 3.x installed on your machine.
- No external libraries are required (uses standard `sys`, `os`, and `re`).

3. HOW TO USE THE SCRIPT

3.1. Command Line Syntax

Open your terminal or command prompt and run the script following this pattern:

```
python bin2c.py <file1.bin> <file2.bin> ... <fileN.bin>
```

3.2. Example Usage

To convert three image files (`test_A.bin`, `test_B.bin`, and `noise.bin`) into a single header file:

```
python bin2c.py test_A.bin test_B.bin noise.bin
```

3.3. Output

The script generates a file named `images_data.h` in the same directory.

- If the file already exists, it will be overwritten.
- The script automatically sanitises filenames to create valid C variable names (e.g., `test-image.bin` becomes `test_image_bin`).

3.4. Understanding the Generated Header (`images_data.h`)

The generated header file contains three main sections:

1. Global Definitions: Macros for image dimensions.
2. Data Arrays: The raw bytes of your images stored as `const uint8` arrays.
3. The AFC Bank: An array of pointers allowing you to iterate over all images programmatically.

3.5. File Structure Example

```
#ifndef IMAGES_DATA_H
#define IMAGES_DATA_H

#define IMG_WIDTH 1020
#define IMG_HEIGHT 1020
#define IMG_SIZE (IMG_WIDTH * IMG_HEIGHT)

// 1. Data Arrays
const uint8 test_A_bin[1040400] = { 0xFF, 0x0A, ... };
const uint8 test_B_bin[1040400] = { 0x00, 0x01, ... };

// 2. Number of images
#define AFC_TEST_IMAGES 2

// 3. The Bank Array (array of pointers)
const uint8* const afc_images_bank[AFC_TEST_IMAGES] = {
    test_A_bin,
    test_B_bin
};

#endif // IMAGES_DATA_H
```